

CSCE 670 INFORMATION RETRIEVAL · MODULE 03 · LECTURE NOTES

Ranking

Scoring functions, derived rather than asserted.

MODULE OUTCOMES

- Explain the vector space model and its weighting.
- Derive the components of TF-IDF from stated intuitions.
- Explain BM25 and what each parameter controls.
- Explain the language modelling approach.
- Tune and compare scoring functions.

1 Three intuitions

TERM FREQUENCY, DOCUMENT FREQUENCY, AND LENGTH: THREE INTUITIONS THAT DETERMINE THE FORMULA

A document mentioning the query term more often is more likely to be relevant — but not proportionally so, since the tenth mention adds far less evidence than the second did. **So the term frequency contribution must saturate.**

A term occurring in few documents is more discriminating — 'the' tells you essentially nothing about which document you want and 'parvovirus' tells you a great deal. **So the weight should be inverse in document frequency.**

And a long document matches more query terms by chance alone — so there must be some normalisation by length — **though a long document may also genuinely cover more material, which is why the normalisation should be partial rather than complete** (§3's b parameter).

Every scoring function in this module is a different way of combining those three intuitions — which is why they all perform similarly, and why understanding the intuitions matters considerably more than memorising any particular formula.

$$w(t,d) = (1 + \log tf(t,d)) \cdot \log(N / df(t))$$

- **Log-damped term frequency times inverse document frequency.** The logarithm on tf is the saturation from §1's first intuition, and the logarithm on N/df keeps the IDF from dominating entirely.
- **The document score is the cosine similarity between the query and document vectors** — which is where the length normalisation enters, as the division by the vector norms.
- **And the cosine normalisation over-penalises long documents — which is the specific, measured limitation that BM25 addresses with a tunable normalisation** (§3).

- **Naming the specific defect makes BM25's design intelligible** rather than arbitrary: it is not a different idea but a correction to one parameter of this one, arrived at by measurement.

2 The vector space model

- **Documents and queries as vectors over the vocabulary**, with cosine similarity as the relevance score — which is intuitive, cheap, and gives a total ordering over the collection.
- **And it handles partial matches naturally**, which Boolean retrieval cannot — **a document containing three of four query terms receives a score rather than a rejection**, which was the advance that made ranked retrieval possible at all.
- **But the geometry is a convenience rather than a justification. Nothing says relevance is a cosine**, and **the model has no probabilistic interpretation** — so there is no principled way to extend it or to incorporate other evidence.
- **And it assumes term independence**, which is plainly false of language (CSCE 638 Module 01 §1's compositionality) — **though it is false in a way that happens not to hurt retrieval much**, which is worth noting honestly rather than pretending the assumption holds.
- **So it works well and explains nothing** — which is a **fair summary**, and **the dissatisfaction with that is exactly what produced the probabilistic models** of §3 and §4. The field's progress here came from wanting a derivation rather than from wanting better numbers.

3 BM25

$$\text{score} = \sum \text{IDF}(t) \cdot \text{tf} \cdot (k_1 + 1) / (\text{tf} + k_1 \cdot (1 - b + b \cdot |d| / \text{avgl}))$$

- **Saturating term frequency, inverse document frequency, and a tunable length normalisation** — §1's three intuitions, each with an explicit parameter.
- **k_1 controls the saturation rate. A low k_1 makes the term frequency contribution saturate quickly; $k_1 \rightarrow \infty$ makes it linear.** Typically between 1.2 and 2.0.
- **b controls the length normalisation. $b = 0$ applies none at all; $b = 1$ applies it fully.** Typically around 0.75, **and it is worth tuning per collection** because the right amount depends on whether long documents in your collection genuinely cover more.
- **The tunable b is the advance over the cosine's fixed normalisation** — it lets the collection decide how much document length should matter, rather than imposing a single answer. **Both parameters should be tuned rather than defaulted**, and the tuning is a grid search over a held-out query set.

BM25 IS A STRONG BASELINE THAT A DECADE OF NEURAL WORK STRUGGLED TO BEAT

It is cheap to compute, requires no training data at all, and transfers across collections and domains without any adaptation — three properties that no learned method has, and which make it the sensible default for a new collection.

And the replication record is pointed: several waves of reported neural retrieval improvements did not survive comparison against a properly tuned BM25, particularly in the zero-shot and out-of-domain settings — which is CSCE 676 Module 13's baseline problem arriving in this field and is documented in the BEIR results.

The operative phrase is 'properly tuned': a BM25 run with library default parameters on a collection it was not tuned for is a weak baseline, and comparisons against it are not informative — which is how a good deal of the overstated literature was produced.

So tune it, report its parameters, and treat beating it as the bar — which is Module 05 §4's requirement and Project 2's third criterion.

4 Language models, and what to do

QUERY LIKELIHOOD: RANK DOCUMENTS BY THE PROBABILITY THEIR LANGUAGE MODEL GENERATES THE QUERY

Estimate a language model from each document, then score each document by $P(\text{query} \mid \text{that document's model})$ — which is a clean derivation from an explicitly stated generative assumption, and that is the point: the formula follows rather than being asserted.

And the smoothing is where all the work happens. An unsmoothed document model assigns probability zero to any query term the document does not contain, which makes the whole score zero — so the collection-wide model is mixed in, by Dirichlet smoothing or Jelinek–Mercer interpolation.

Which turns out to play exactly the role IDF plays: the smoothing makes common terms contribute less to the score, because they are already well explained by the collection model and therefore carry little document-specific evidence. IDF emerges from the smoothing rather than being postulated, which is the satisfying part of this derivation.

So two independent derivations arrive at similar behaviour — which is evidence that §1's three intuitions are the actual content, and the formula is the packaging, and it is why the practical advice in this module is to pick one and move on.

- Use **BM25, tuned on your own collection**, as the baseline and frequently as the production system — it is still what many deployed search systems run.
- **Score fields separately and combine them** — a term appearing in the title means considerably more than the same term in the body, and **BM25F is the standard extension that handles this** without simply concatenating the fields (which breaks the length normalisation).
- **Tune k_1 and b by grid search** on a held-out query set, and **report the values you used**, because **an untuned comparison is uninformative** and a reader cannot assess it.
- **And compare against query likelihood once**, to see for yourself how similar they are — **which calibrates your expectations about how much the scoring function choice matters** before you spend time on it.
- **Then spend your remaining effort on Modules 06 and 07**, because **the scoring function is not where the headroom is** — and **that is a useful thing to establish early, by measuring rather than by assumption**, which is what the comparison above is for.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapters 6, 11, and 12 (free)	§1 through §4, with the probabilistic derivations done carefully.

Resource	What it covers for this module
Robertson & Zaragoza — The Probabilistic Relevance Framework (free)	§3 in the original , by its authors — including where BM25 came from and what it approximates.
Zhai & Lafferty — A study of smoothing methods (free)	§4's smoothing, measured — and the IDF-emerges-from-smoothing argument.
Thakur et al. — BEIR (free)	§3's callout, with the numbers — BM25 against neural methods across eighteen zero-shot collections.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the three intuitions** and say which parameter of BM25 corresponds to each.
- 2 **Implement TF-IDF with cosine** and inspect the top ten for five queries.
- 3 **Find a case where the cosine over-penalises a long document.**
- 4 **Implement BM25 from the formula** and verify against a reference implementation.
- 5 **Sweep k_1 from 0.5 to 5** and plot the effect on your metric.
- 6 **Sweep b from 0 to 1** and explain the shape of the curve.
- 7 **Compare default and tuned BM25** and report the gap.
- 8 **Implement query likelihood with Dirichlet smoothing** and compare against BM25.
- 9 **Show that the smoothing weight behaves like IDF** on a common and a rare term.
- 10 **Implement BM25F over title and body** and measure the improvement.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the three intuitions and why each is as it is.
- 2 Why must term frequency saturate, and why is length normalisation partial?
- 3 Give TF-IDF and say where the length normalisation comes from.
- 4 What is the cosine's specific defect?
- 5 What does the vector space model give, and what are its three limits?
- 6 Give BM25 and say what k_1 and b each control.
- 7 Why is BM25 a strong baseline, and what does 'properly tuned' mean?

- 8 Describe query likelihood and why smoothing is necessary.
- 9 How does IDF emerge from the smoothing?
- 10 What does the agreement of two derivations suggest?